Supplemental Material: What static transport data determine about a conserving phonon closure

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[PENDING: content of Secs. S1, S3 and S5–S11]
Numbers of equations, statements and hypotheses without the prefix S refer to the main text.

S1. EVENT DECOMPOSITION OF THE THREE-PHONON OPERATOR AND ENERGY-CONSERVING STOICHIOMETRY

S2. LIFETIMES AND DC RESPONSE: SHARPNESS OF THE BOUND $K_0 \geq K_{\mathrm{RTA}}/\varsigma_{\max}$ AND SELF-ENERGY LINEWIDTHS

Equality networks. By Lemma 2, the currents that attain $K_0 = K_{\mathrm{RTA}}/\varsigma_{\max}$ are the vectors $\mathbf{j} = \mathcal{C}\mathbf{y}$ with $\mathbf{y} \neq 0$ and $\mathcal{C}\mathbf{y} = \varsigma_{\max}\,\mathrm{diag}(\mathbf{r})\,\mathbf{y}$; this requires that every active event touch exactly $\varsigma_{\max}$ modes and that the numbers $\zeta_{\gamma\nu}y_\nu$ be equal on the support of each active event. Conversely, if both conditions hold for some $\mathbf{y} \neq 0$, with common value c_γ on $\mathrm{supp}\,\boldsymbol{\zeta}_\gamma$, then $\boldsymbol{\zeta}_\gamma^{\mathsf{T}}\mathbf{y} = \varsigma_{\max}c_\gamma$ and $\zeta_{\gamma\nu}^2 y_\nu = \zeta_{\gamma\nu}c_\gamma$, so that

$$\mathcal{C}\mathbf{y} = \sum_\gamma g_\gamma\,\boldsymbol{\zeta}_\gamma\,(\boldsymbol{\zeta}_\gamma^{\mathsf{T}}\mathbf{y}) = \varsigma_{\max}\sum_\gamma g_\gamma c_\gamma\,\boldsymbol{\zeta}_\gamma = \varsigma_{\max}\,\mathrm{diag}(\mathbf{r})\,\mathbf{y}, \tag{S1}$$

because $r_\nu y_\nu = \sum_\gamma g_\gamma \zeta_{\gamma\nu}^2 y_\nu$. Then $\mathbf{j} = \mathcal{C}\mathbf{y} \in \mathcal{H}\setminus\{0\}$, $K_0 = \mathbf{y}^{\mathsf{T}}\mathcal{C}\mathbf{y} = \varsigma_{\max}\,\mathbf{y}^{\mathsf{T}}\mathrm{diag}(\mathbf{r})\,\mathbf{y}$ and $K_{\mathrm{RTA}} = \varsigma_{\max}^2\,\mathbf{y}^{\mathsf{T}}\mathrm{diag}(\mathbf{r})\,\mathbf{y}$, so that $K_0/K_{\mathrm{RTA}} = 1/\varsigma_{\max}$ exactly.

For event vectors $\boldsymbol{\zeta}_\gamma = \mathsf{W}^{-1}\mathbf{s}_\gamma$ [Eq. (15), derived there for three-phonon events; the same linearization gives $\boldsymbol{\zeta}_\gamma = \mathsf{W}^{-1}\mathbf{s}_\gamma$ for $1 \leftrightarrow 3$ and $2 \leftrightarrow 2$ events] and $\mathbf{y} = \mathsf{W}\boldsymbol{\psi}$, one has $\zeta_{\gamma\nu}y_\nu = s_{\gamma\nu}\psi_\nu$, so the conditions involve the stoichiometry only. If every event of a network touches $\varsigma_{\max}$ modes and $s_{\gamma\nu}\psi_\nu$ takes a single value on $\mathrm{supp}\,\mathbf{s}_\gamma$ for every γ, then Eq. (S1) holds for all positive weights w_ν and all positive rates g_γ, and the current $\mathbf{j} = \mathcal{C}\,\mathsf{W}\boldsymbol{\psi}$ attains $K_0/K_{\mathrm{RTA}} = 1/\varsigma_{\max}$. Table S1 lists three such networks. In each, the events conserve energy ($\mathbf{s}_\gamma^{\mathsf{T}}\boldsymbol{\varepsilon} = 0$) and the stoichiometric vectors have rank $n-1$; since W is invertible, the $\boldsymbol{\zeta}_\gamma$ then span $\mathcal{H}$, and every positive rate vector is admissible. In the chain, $\psi_{\nu+1} = -2\psi_\nu$ gives $s_{\gamma\mu}\psi_\mu = 2\psi_\nu$ on both modes $\mu = \nu, \nu+1$ of the event $(\nu+1) \to \nu+\nu$. In the other two networks no mode is both a parent and a daughter, and $\psi = +1$ on parents and -1 on daughters gives $s_{\gamma\nu}\psi_\nu = -1$ on every support; the events of the network with $\varsigma_{\max} = 4$ are $1 \leftrightarrow 3$ four-phonon processes. A network of $2 \leftrightarrow 2$ events alone conserves the phonon number ($\mathbf{s}_\gamma^{\mathsf{T}}\mathbf{1} = 0$), an invariant other than the energy unless all modes have the same energy, and is then not admissible.

TABLE S1. Networks that attain $K_0 = K_{\mathrm{RTA}}/\varsigma_{\max}$ for all positive weights and rates. Modes are numbered $1, \ldots, n$; $\mathsf{p} \to \mathsf{a}+\mathsf{b}$ denotes the event with stoichiometric vector $-\boldsymbol{\delta}_\mathsf{p} + \boldsymbol{\delta}_\mathsf{a} + \boldsymbol{\delta}_\mathsf{b}$ ($-\boldsymbol{\delta}_\mathsf{p} + 2\boldsymbol{\delta}_\mathsf{a}$ if $\mathsf{a} = \mathsf{b}$), and similarly for three daughters. The current is $\mathbf{j} = \mathcal{C}\,\mathsf{W}\boldsymbol{\psi}$.

$\varsigma_{\max}$	n; energies ε_ν	events	$\boldsymbol{\psi}$	K_0/K_{RTA}
2	5; $(1, 2, 4, 8, 16)$	$2 \to 1+1,\ 3 \to 2+2,\ 4 \to 3+3,\ 5 \to 4+4$	$(1, -2, 4, -8, 16)$	1/2
3	8; $(6, 6, 6, 3, 3, 3, 3, 3)$	$1 \to 4+5,\ 1 \to 4+6,\ 1 \to 4+7,\ 1 \to 4+8,$ $1 \to 5+6,\ 2 \to 4+5,\ 3 \to 4+5$	$(1, 1, 1, -1, \ldots, -1)$	1/3
4	9; $(6, 6, 6, 2, 2, 2, 2, 2, 2)$	$1 \to 4+5+6,\ 1 \to 4+5+7,\ 1 \to 4+5+8,$ $1 \to 4+5+9,\ 1 \to 4+6+7,\ 1 \to 5+6+7,$ $2 \to 4+5+6,\ 3 \to 4+5+6$	$(1, 1, 1, -1, \ldots, -1)$	1/4

Verification. For each network of Table S1, the script `checks/verify_manuscript_algebra.py` of the manuscript sources (block "Lemma 2 sharpness") checks in exact arithmetic the energy conservation, the rank $n-1$, the identity $\mathcal{C}\mathbf{y} = \varsigma_{\max}\,\mathrm{diag}(\mathbf{r})\,\mathbf{y}$ for symbolic weights and rates, and the ratio $K_0/K_{\mathrm{RTA}} = 1/\varsigma_{\max}$ for rational weights and unequal

rates, with $K_0 = \mathbf{j}^{\mathsf{T}}(\mathcal{C} + \mathbf{e}\mathbf{e}^{\mathsf{T}}/|\mathbf{e}|^2)^{-1}\mathbf{j}$ computed without using Eq. (S1). The script `checks/audit/audit_lemma2.py` checks the same networks with other weights and rates.

Networks with events of different sizes. By Lemma 2 the inequality is strict when an active event touches fewer than $\varsigma_{\max}$ modes; the constant can nevertheless not be improved. Writing $\mathbf{j} = \operatorname{diag}(\mathbf{r})^{1/2}\mathbf{u}$, where $\mathbf{u}$ lies in the range of $G = \operatorname{diag}(\mathbf{r})^{-1/2}\,\mathcal{C}\,\operatorname{diag}(\mathbf{r})^{-1/2}$ exactly when $\mathbf{j} \in \mathcal{H}$, gives $K_0 = \mathbf{u}^{\mathsf{T}}G^{+}\mathbf{u}$ (if $G\mathbf{z} = \mathbf{u}$, then $\mathcal{C}\operatorname{diag}(\mathbf{r})^{-1/2}\mathbf{z} = \mathbf{j}$) and $K_{\mathrm{RTA}} = |\mathbf{u}|^2$. Hence $\min_{\mathbf{j}\in\mathcal{H}\setminus\{0\}} K_0/K_{\mathrm{RTA}} = 1/\lambda_{\max}(G)$, and $\lambda_{\max}(G)$ is at least $\mathbf{y}^{\mathsf{T}}\mathcal{C}\mathbf{y}/\mathbf{y}^{\mathsf{T}}\operatorname{diag}(\mathbf{r})\,\mathbf{y}$ for every $\mathbf{y} \neq 0$. Take six modes with energies $(1,2,3,4,5,6)$ and unit weights, the $2 \leftrightarrow 2$ event $2+5 \to 3+4$ with rate 1, and the six three-phonon events $3 \to 1+2$, $4 \to 1+3$, $5 \to 1+4$, $5 \to 2+3$, $6 \to 1+5$ and $6 \to 2+4$ with rate $\delta > 0$. Then $\varsigma_{\max} = 4$, the stoichiometric rank is 5, so that the rates are admissible, and $\mathbf{y} = (0,1,-1,-1,1,0)^{\mathsf{T}}$ gives $\mathbf{y}^{\mathsf{T}}\mathcal{C}\mathbf{y} = 16 + 10\delta$ and $\mathbf{y}^{\mathsf{T}}\operatorname{diag}(\mathbf{r})\,\mathbf{y} = 4 + 12\delta$. Therefore

$$1 < \min_{\mathbf{j}\in\mathcal{H}\setminus\{0\}} \frac{4K_0}{K_{\mathrm{RTA}}} \leq \frac{16+48\delta}{16+10\delta} \to 1 \qquad (\delta \to 0). \tag{S2}$$

The minima $4/\lambda_{\max}(G)$ are 1.2198, 1.0234, 1.00236 and 1.000024 for $\delta = 10^{-1}$, 10^{-2}, 10^{-3} and 10^{-5} (script `checks/audit/audit_lemma2.py`).

Self-energy linewidths. Let $\tilde{\mathbf{r}} \leq \mathbf{r}$ be a linewidth that is strictly smaller than $\mathbf{r}$ on at least one mode carrying current, such as one that counts the self-coupling of a repeated daughter with half weight. At an equality current $\mathbf{j} = \mathcal{C}\mathbf{y}$ of Lemma 2, $K_0 = K_{\mathrm{RTA}}/\varsigma < \tilde{K}_{\mathrm{RTA}}/\varsigma$, so the inequality fails for $\tilde{\mathbf{r}}$. For the chain with energies $1, 2, 4, 8, 16$ ($\varsigma = 2$) and the half-weight self-coupling, $K_0/(\tilde{K}_{\mathrm{RTA}}/2) = 0.722$, 0.686 and 0.688 for three random sets of weights and rates, in exact arithmetic (`checks/audit_supp/audit_s2_networks.py`).

S3. LIFETIME-CONSTRAINED SHARPNESS OF THE CONE BOUND: AN EXACT SEVEN-MODE EXAMPLE

S4. INFRARED CRITERION: LATTICE SUMS AND MESH LAWS

This section proves the mesh laws of Theorem 3(i) and the lower bound used for Theorem 3(ii); the integral criterion is proved in the main text. The meshes are $\mathbf{k} = \mathsf{B}\mathbf{n}/N_1$ with $\mathbf{n} \in \mathbb{Z}^d$, where the columns of B are primitive reciprocal-lattice vectors, so that $|\det \mathsf{B}|\, V_{\mathrm{cell}} = (2\pi)^d$ and $\mathcal{V} = N_1^d V_{\mathrm{cell}}$. The radius k_0 is taken so small that the ball $k < k_0$ lies inside the Brillouin zone and the $o(1)$ of (H1) is at most $1/2$ there. As in (H2) and (H5), $\Phi_m = \sum_{\mathrm{ac}} J_0^2 r_0^{-m}$ and $A_m = \int_{S^{d-1}} \Phi_m \, d\Omega$; the mesh moments are $\mathcal{M}_m(N_1) = \mathcal{V}^{-1}\sum_\nu J(\mathbf{k}_\nu, s_\nu)^2 r_\nu^{-m}$.

Lemma S1 (Lattice sums)**.** *Let $\mathsf{B} \in \mathbb{R}^{d\times d}$ be invertible, $m \geq 0$ and $\alpha > 0$, and let $\Phi\colon S^{d-1} \to [0,\infty)$ be bounded and continuous outside a closed set of zero measure, with $A = \int_{S^{d-1}} \Phi \, d\Omega > 0$. For $\mathbf{n} \in \mathbb{Z}^d \setminus \{0\}$ put $\varphi(\mathbf{n}) = \Phi(\mathsf{B}\mathbf{n}/|\mathsf{B}\mathbf{n}|)\,|\mathsf{B}\mathbf{n}|^{-m\alpha}$. Let $k_0 > 0$, let ϑ be a bounded function on $(0,k_0)$ with $\vartheta(t) \to 0$ as $t \to 0$, and for $N \in \mathbb{N}$ let $\epsilon_N(\mathbf{n})$ be real numbers with $|\epsilon_N(\mathbf{n})| \leq \vartheta(|\mathsf{B}\mathbf{n}|/N)$. Put*

$$S_N = \sum_{0<|\mathsf{B}\mathbf{n}|<k_0 N} \big[\varphi(\mathbf{n}) + \epsilon_N(\mathbf{n})\,|\mathsf{B}\mathbf{n}|^{-m\alpha}\big], \qquad \bar{\vartheta} = \sup_{0<t<k_0} \vartheta(t). \tag{S3}$$

(a) If $m\alpha < d$, there is C, depending only on B, d and $m\alpha$, such that $N^{m\alpha-d}\sum_{0<|\mathsf{B}\mathbf{n}|<\delta N} |\mathsf{B}\mathbf{n}|^{-m\alpha} \leq C\delta^{d-m\alpha}$ for all N and all $\delta > 0$, and

$$\limsup_{N\to\infty} \Big| N^{m\alpha-d} S_N - \frac{A\,k_0^{d-m\alpha}}{(d-m\alpha)\,|\det \mathsf{B}|} \Big| \leq \frac{\bar{\vartheta}\,|S^{d-1}|\,k_0^{d-m\alpha}}{(d-m\alpha)\,|\det \mathsf{B}|}. \tag{S4}$$

(b) If $m\alpha = d$, then $S_N = |\det \mathsf{B}|^{-1} A \log N + o(\log N)$.

(c) If $m\alpha > d$, then $S_N \to S_\Phi := \sum_{\mathbf{n}\neq 0}\varphi(\mathbf{n})$, and $0 < S_\Phi < \infty$.

Proof. Counting. Let $\rho_0 = \min_{\mathbf{n}\neq 0}|\mathsf{B}\mathbf{n}| > 0$. The balls of radius $\rho_0/2$ centred at the points $\mathsf{B}\mathbf{n}$ are disjoint, so for $R \geq \rho_0$ at most $(3R/\rho_0)^d$ of these points satisfy $0 < |\mathsf{B}\mathbf{n}| < R$. The dyadic shell $\Sigma_l = \{\mathbf{n} : 2^l \leq |\mathsf{B}\mathbf{n}| < 2^{l+1}\}$ ($l \in \mathbb{Z}$) is empty unless $2^{l+1} > \rho_0$, and then

$$\sum_{\mathbf{n}\in\Sigma_l} |\mathsf{B}\mathbf{n}|^{-m\alpha} \leq C_1\, 2^{l(d-m\alpha)}, \qquad C_1 = (6/\rho_0)^d. \tag{S5}$$

Shell Riemann sums. Let $D = \{\mathbf{u} \in \mathbb{R}^d : 1 \le |\mathsf{B}\mathbf{u}| < 2\}$ and $\tilde\varphi(\mathbf{u}) = \Phi(\mathsf{B}\mathbf{u}/|\mathsf{B}\mathbf{u}|)\,|\mathsf{B}\mathbf{u}|^{-m\alpha}$. On D, $\tilde\varphi$ is bounded and continuous outside a set of zero measure (a cone over the discontinuity set of Φ), and the boundary of D, two ellipsoids, has zero measure. By Lebesgue's criterion $\tilde\varphi$ is Riemann integrable on D, so the lattice sums $h^d \sum_{\mathbf{u}\in h\mathbb{Z}^d\cap D} \tilde\varphi(\mathbf{u})$, which are Riemann sums for the partition of space into the cubes $\mathbf{u} + [0,h)^d$, converge as $h \to 0$ to $c_* = \int_D \tilde\varphi\, d\mathbf{u} = |\det \mathsf{B}|^{-1} A \int_1^2 \rho^{d-1-m\alpha} d\rho > 0$; the last form follows from the substitution $\mathbf{v} = \mathsf{B}\mathbf{u}$ and polar coordinates. Since $\varphi(\mathbf{n}) = 2^{-lm\alpha}\tilde\varphi(2^{-l}\mathbf{n})$, and $\mathbf{n} \in \Sigma_l$ if and only if $2^{-l}\mathbf{n} \in D$,

$$a_l := \sum_{\mathbf{n}\in\Sigma_l} \varphi(\mathbf{n}) = 2^{l(d-m\alpha)}\Big[2^{-ld} \sum_{\mathbf{u}\in 2^{-l}\mathbb{Z}^d\cap D} \tilde\varphi(\mathbf{u})\Big] = 2^{l(d-m\alpha)}\,[c_* + o(1)] \qquad (l \to \infty). \tag{S6}$$

Part (a). Summing Eq. (S5) over the shells with $2^l < R$ gives $\sum_{0<|\mathsf{B}\mathbf{n}|<R} |\mathsf{B}\mathbf{n}|^{-m\alpha} \le C_1 R^{d-m\alpha}/(1-2^{m\alpha-d})$ for every $R > 0$, which is the first claim with $R = \delta N$. For $0 < \delta < k_0$, $N^{m\alpha-d}\sum_{\delta N\le|\mathsf{B}\mathbf{n}|<k_0 N} \varphi(\mathbf{n}) = N^{-d}\sum_{\mathbf{u}\in N^{-1}\mathbb{Z}^d,\,\delta\le|\mathsf{B}\mathbf{u}|<k_0} \tilde\varphi(\mathbf{u})$ is a Riemann sum of $\tilde\varphi$ over $\{\delta \le |\mathsf{B}\mathbf{u}| < k_0\}$, on which $\tilde\varphi$ is Riemann integrable as above; it converges to $|\det \mathsf{B}|^{-1} A\,(k_0^{d-m\alpha} - \delta^{d-m\alpha})/(d-m\alpha)$. The points with $0 < |\mathsf{B}\mathbf{n}| < \delta N$ contribute at most $C\delta^{d-m\alpha}\sup\Phi$. Letting $N \to \infty$ and then $\delta \to 0$ gives the limit of $N^{m\alpha-d}\sum_{0<|\mathsf{B}\mathbf{n}|<k_0 N}\varphi(\mathbf{n})$ displayed in Eq. (S4). The same argument with $\Phi \equiv 1$ gives the limit of $N^{m\alpha-d}\sum_{0<|\mathsf{B}\mathbf{n}|<k_0 N}|\mathsf{B}\mathbf{n}|^{-m\alpha}$, and the error terms are at most $\bar\vartheta$ times this sum.

Part (b). Now $2^{l(d-m\alpha)} = 1$ and $a_l \to c_* = |\det \mathsf{B}|^{-1} A \log 2$. Let L_N be the integer with $2^{L_N} \le k_0 N < 2^{L_N+1}$. The points with $0 < |\mathsf{B}\mathbf{n}| < 2^{L_N}$ contribute $\sum_{l<L_N} a_l = c_* L_N + o(L_N)$, by Eq. (S6) and because only finitely many shells with $l < 0$ are nonempty; the remaining points lie in Σ_{L_N} and contribute at most $C_1 \sup\Phi$. Since $L_N \log 2 = \log N + O(1)$, the sum of φ is $|\det \mathsf{B}|^{-1} A \log N + o(\log N)$. For the error terms, given $\epsilon' > 0$ choose $\delta \in (0, k_0)$ with $\vartheta(t) \le \epsilon'$ for $t < \delta$. By Eq. (S5), the points with $|\mathsf{B}\mathbf{n}| < \delta N$ contribute at most $\epsilon' C_1$ times the number of nonempty shells below δN, which is $\log_2 N + O(1)$, and the points with $\delta N \le |\mathsf{B}\mathbf{n}| < k_0 N$ at most $\bar\vartheta C_1[\log_2(k_0/\delta) + 2]$, which does not depend on N. Since ϵ' is arbitrary, the error terms are $o(\log N)$.

Part (c). By Eq. (S5), $\sum_{\mathbf{n}\ne 0} |\mathsf{B}\mathbf{n}|^{-m\alpha} \le C_1 \sum_{l:\,2^{l+1}>\rho_0} 2^{l(d-m\alpha)} < \infty$, so S_Φ converges and the truncated sums of φ converge to it. The error terms are dominated by $\bar\vartheta\,|\mathsf{B}\mathbf{n}|^{-m\alpha}$ and tend to zero for every fixed $\mathbf{n}$ as $N \to \infty$, because $|\mathsf{B}\mathbf{n}|/N \to 0$; by dominated convergence their sum tends to zero. Finally, $\{\Phi > 0\}$ has positive measure and the discontinuity set of Φ has zero measure, so Φ is continuous and positive at some direction and positive on a neighbourhood U of it. The directions $\mathsf{B}\mathbf{n}/|\mathsf{B}\mathbf{n}|$ are dense in S^{d-1}, being the image of the rational directions under a homeomorphism of the sphere; some $\mathbf{n}$ has $\mathsf{B}\mathbf{n}/|\mathsf{B}\mathbf{n}| \in U$, and then $\varphi(\mathbf{n}) > 0$. □

Regularity is necessary. Let $d \ge 2$. If $\Phi = 0$ on the countable set of directions $\mathsf{B}\mathbf{n}/|\mathsf{B}\mathbf{n}|$ and $\Phi = 1$ elsewhere, then $A = |S^{d-1}| > 0$ but $\varphi \equiv 0$, and S_N consists of the error terms alone. Taking $J = J_0(\hat{\mathbf{k}})$ with such a J_0^2 and $r = k^\alpha$ near Γ satisfies (H1)–(H3), but every mesh point near Γ then has $J = 0$ and the mesh laws fail. Likewise, setting $J = 0$ at the mesh points with $k \ge k_0$ of all meshes, a countable set, changes neither (H1)–(H3) nor any $\mathcal{M}_m$, but removes the part $k \ge k_0$ of every mesh sum. Hypothesis (H5) excludes both constructions and admits the discontinuities of frequency-sorted acoustic branches, which lie on a closed set of directions of zero measure (acoustic axes, where branches touch conically and the group velocity of a sorted branch depends on the direction of approach, and curves where branches cross).

Mesh laws. Assume (H1)–(H3) and (H5). For $k < k_0$, (H1)–(H2) give $\sum_{\mathrm{ac}} J^2 r^{-m} = [\Phi_m(\hat{\mathbf{k}}) + \epsilon_{\mathrm{ac}}(\mathbf{k})]\,k^{-m\alpha}$ with $\sup_{\hat{\mathbf{k}}} |\epsilon_{\mathrm{ac}}(\mathbf{k})| \to 0$ as $k \to 0$; the error is additive because J_0 may vanish on a plane. At $\mathbf{k} = \mathsf{B}\mathbf{n}/N_1$ the acoustic part $k_\nu < k_0$ of $\mathcal{M}_m(N_1)$ is therefore $N_1^{m\alpha-d} V_{\mathrm{cell}}^{-1} S_{N_1}$, with S_N of Eq. (S3) for $\Phi = \Phi_m$ and $\epsilon_N(\mathbf{n}) = \epsilon_{\mathrm{ac}}(\mathsf{B}\mathbf{n}/N)$, while the optical branches and the part $k \ge k_0$ contribute a bounded amount, by (H3) and the boundedness of J. Lemma S1 and $|\det \mathsf{B}|\,V_{\mathrm{cell}} = (2\pi)^d$ give:

- $m\alpha > d$: $\mathcal{M}_m(N_1) = N_1^{m\alpha-d} S_m/V_{\mathrm{cell}}\,[1 + o(1)]$, with $S_m = S_{\Phi_m} \in (0,\infty)$;
- $m\alpha = d$: $\mathcal{M}_m(N_1) = (2\pi)^{-d} A_m \log N_1 + o(\log N_1)$;
- $m\alpha < d$: for $0 < \delta < k_0$ the part $k < \delta$ of $\mathcal{M}_m(N_1)$ is at most $C'\delta^{d-m\alpha}$ for all N_1, by Lemma S1(a) for the acoustic branches and by the same lemma with $m = 0$ for the optical branches (at most $C''\delta^d$), and so is the part $k < \delta$ of $\mathcal{M}_m$. On $k \ge \delta$ the integrands are bounded, by (H1) and (H3), and Riemann integrable, by (H5), so the remaining mesh sums converge to the remaining part of $\mathcal{M}_m$. Hence $\limsup_{N_1\to\infty} |\mathcal{M}_m(N_1) - \mathcal{M}_m| \le 2C'\delta^{d-m\alpha}$ for every δ, that is, $\mathcal{M}_m(N_1) \to \mathcal{M}_m$.

For $\tau_{\mathrm{mem}}^{\mathrm{RTA}} = \mathcal{M}_2(N_1)/\mathcal{M}_1(N_1)$ on the meshes this gives:

- $2\alpha < d$: $\tau_{\mathrm{mem}}^{\mathrm{RTA}} \to \mathcal{M}_2/\mathcal{M}_1$;

- $2\alpha = d$: $\tau_{\mathrm{mem}}^{\mathrm{RTA}} = (2\pi)^{-d} A_2 \log N_1 / \mathcal{M}_1 \, [1 + o(1)]$;
- $d/2 < \alpha < d$: $\tau_{\mathrm{mem}}^{\mathrm{RTA}} = N_1^{2\alpha-d} S_2 / (V_{\mathrm{cell}} \mathcal{M}_1) \, [1 + o(1)]$;
- $\alpha = d$: $\tau_{\mathrm{mem}}^{\mathrm{RTA}} = (2\pi)^d S_2 N_1^d / (V_{\mathrm{cell}} A_1 \log N_1) \, [1 + o(1)]$.

Event operator. Under the uniform (H1) of Theorem 3(ii) there is $\delta_1(k) \to 0$, independent of N_1, with $r_\nu \le r_0(\hat{\mathbf{k}}_\nu) k_\nu^\alpha [1 + \delta_1(k_\nu)]$ for all N_1 and all acoustic modes with $k_\nu < k_0$. Hence

$$\sum_{\mathrm{ac},\, k_\nu < k_0} j_\nu^2 r_\nu^{-2} \;\ge\; \frac{N_1^{2\alpha-d}}{V_{\mathrm{cell}}} \sum_{0 < |\mathsf{B}\mathbf{n}| < k_0 N_1} \big[\varphi(\mathbf{n}) + \epsilon'(\mathsf{B}\mathbf{n}/N_1) \, |\mathsf{B}\mathbf{n}|^{-2\alpha} \big], \tag{S7}$$

where φ is built from Φ_2 and $\epsilon'(\mathbf{k}) = \sum_{\mathrm{ac}} J^2 r_0^{-2} [1+\delta_1(k)]^{-2} - \Phi_2(\hat{\mathbf{k}})$ depends on $\mathbf{k}$ only and tends to zero uniformly in $\hat{\mathbf{k}}$. By Lemma S1(b) and (c) the right side equals $(2\pi)^{-d} A_2 \log N_1 \, [1+o(1)]$ for $2\alpha = d$ and $N_1^{2\alpha-d} S_2 / V_{\mathrm{cell}} \, [1+o(1)]$ for $2\alpha > d$. Only the continuity of Φ_2 outside a closed set of zero measure enters, not the Riemann integrability of $J^2 r^{-2}$ away from Γ, which for $\mathbf{r} = \operatorname{diag} \mathcal{C}_{N_1}$ would depend on N_1. With (H4) and Proposition 2 this proves Theorem 3(ii).

Numerical check. The script `checks/audit/audit_ir_lattice.py` of the manuscript sources compares the constants with direct lattice sums (double precision; continuum moments and A_m by adaptive quadrature; S_2 by a lattice sum over $|\mathbf{n}| \le 1500$ with a continuum tail). The test uses $d = 2$, $\mathsf{B} = 2\pi I$ ($V_{\mathrm{cell}} = 1$), odd N_1 from 101 to 3201, $J^2 = (\hat{k}_x^2 + 0.3k)\, e^{-k^2/8}$ and $r = (1 + 2.4 \hat{k}_x^2 \hat{k}_y^2)\, k^\alpha (1 + 0.5k^2)$. Here $J_0^2 = \hat{k}_x^2$ vanishes on the line $\hat{k}_x = 0$, so the correction to J^2 is additive. Table S2 gives the results at $N_1 = 3201$. The laws are approached slowly. Through $\mathcal{M}_1(N_1)$, which in this test converges like $N_1^{-(d-\alpha)}$, the ratio of $\tau_{\mathrm{mem}}^{\mathrm{RTA}}$ to $N_1^{2\alpha-d} S_2 / (V_{\mathrm{cell}} \mathcal{M}_1)$ is still 1.026 at $N_1 = 3201$ for $\alpha = 1.5$, and at $\alpha = d$ the relative correction decays like $1/\log N_1$. Exponents fitted on coarse meshes are therefore biased.

TABLE S2. Mesh laws of Theorem 3(i) against direct lattice sums ($d = 2$; model in the text).

α	prediction	result at $N_1 = 3201$
0.6 ($2\alpha < d$)	$\tau_{\mathrm{mem}}^{\mathrm{RTA}} \to \mathcal{M}_2/\mathcal{M}_1 = 0.6291354$	relative error 2.6×10^{-3}, decreasing like $N_1^{-0.79}$ ($d - 2\alpha = 0.8$)
1 ($2\alpha = d$)	slope $(2\pi)^{-2} A_2/\mathcal{M}_1 = 0.3885388$ of $\tau_{\mathrm{mem}}^{\mathrm{RTA}}$ against $\log N_1$	local slope 0.99657 times the prediction, increasing with N_1
1.5 ($d/2 < \alpha < d$)	$\mathcal{M}_2(N_1)/N_1^{2\alpha-d} \to S_2$	relative deviation 2.5×10^{-3}; $\mathcal{M}_1(N_1) \to \mathcal{M}_1$ like $N_1^{-0.500}$ ($d - \alpha = 0.5$); $\tau_{\mathrm{mem}}^{\mathrm{RTA}} / [N_1^{2\alpha-d} S_2/\mathcal{M}_1]$: 0.99785 after removing the $N_1^{-1/2}$ correction of $\mathcal{M}_1$ only, 0.9990 from a three-term fit (proved limit 1)
2 ($\alpha = d$)	slope $(2\pi)^{-2} A_1$ of $\mathcal{M}_1(N_1)$ against $\log N_1$, and $\mathcal{M}_2(N_1)/N_1^d \to S_2$	ratios 1.00087 and 1.00158; $\tau_{\mathrm{mem}}^{\mathrm{RTA}} \log N_1 / N_1^d$ is still 1.057 times its limit

S5. EVENT GEOMETRIES: DEBYE-TYPE MESHES AND ALN

S6. FEASIBLE RANGES OF THE MEMORY MOMENT: OPTIMIZATION, CERTIFIED WITNESSES AND IMPLICIT-FUNCTION TRANSFER

S7. FRONT-FACE QUADRUPOLE: DERIVATION, NUMERICAL STABILITY AND INDEPENDENT CHECKS

S8. RANK THEOREM: PROOF AND COMPUTER-ALGEBRA VERIFICATION

S9. FDTR MODEL: TRANSDUCER, INTERFACES, BEAMS, NUISANCE SETS AND CONVERGENCE

S10. REPRODUCIBILITY: ENVIRONMENT, INPUTS, SCRIPTS AND CHECKSUMS

S11. FAILED APPROACHES THAT AFFECT THE INTERPRETATION